# Pi Coding Agent: A Practical Primer

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*Verified against primary sources for all sections; unverified claims are flagged inline.*

# 1 The Short Version

If you have used Claude Code, Pi still feels sparse on first contact: no built-in plan mode, subagent workflow, per-tool permission prompts, or general web search. But the old claim that it has no Model Context Protocol (MCP) client is now wrong: Pi 1.0 includes built-in support for MCP servers (external tool servers) and codemode for composing tool calls. Pi remains a programmable harness (the program around the model that runs the agent loop)

that you extend. The official documentation is reference-style, and third-party write-ups can describe an earlier package or feature set. This primer is the orientation in between, written for someone who already knows Claude Code.

Pi is a terminal coding agent and an agent toolkit. Its default tools are `read`, `write`, `edit`, and `bash`; optional built-in tools and MCP can expand that set. It provides persistent sessions, a terminal UI, and several ways to add your own behaviour: skills, prompt templates, TypeScript extensions, and packages. You can add a search skill, browser bridge, approval gate, subagent workflow, or local-model provider without forking the program.

Pi was created by Mario Zechner and is MIT-licensed. Since April 2026 the project has been owned by the company Earendil, and the repository lives in the `earendil-works` GitHub organisation. The official changelog records 1.0.0 on 1 October and 1.0.4 on 5 October 2026. This edition uses that release baseline; the official documentation remains authoritative for changing flags and settings.

### 1.1 What You Will Learn

1. Why Pi is built the way it is, and how it maps onto Claude Code.
2. Installing Pi and a controlled first run.
3. Authentication, and hosted, compatible, and local models.
4. Running Pi non-interactively and embedding it.
5. Choosing between a skill, extension, package, or separate service, and a minimal starter setup.
6. Giving Pi a reliable web and forum research path.
7. Sessions: branching, compaction, hand-off, and a working routine.
8. What project trust and containment do, and do not, protect.
9. How Pi compares with OpenCode, Aider, Claude Code, and Codex.
10. A worked data-science project using Pi, Podman, and Rocker.

### 1.2 What This Primer Is Not

This is not a benchmark, a list of every Pi package, or a guarantee that a model, provider, or community extension remains available. It is not a TypeScript API reference.

Pi is not safer than other agents merely because its core is small. It normally runs with the permissions of the launching user. Containment is an operating-system, container, or virtual-machine concern.

You need a terminal, Git, and comfort installing software. Familiarity with Claude Code

(`CLAUDE.md`, skills, slash commands) is assumed. You do not need to know Pi or write TypeScript; the extension section assumes only that you can read a little of it when reviewing code.

# 2 How Pi Works

Before installing anything, it helps to know what Pi is, what it leaves out on purpose, and where a Claude Code habit will or will not carry over. The four subsections below take those in turn.

## 2.1 Pi Is a Harness, Not a Model

An AI coding system has four separable parts:

1. **Model:** predicts text and tool calls. It may be hosted or local.
2. **Harness:** builds prompts, presents tools, sends requests, streams output, and records state.
3. **Tools:** give the model effects such as reading files, editing, running commands, searching, or browsing.
4. **Environment:** supplies permissions, network, filesystem, credentials, operating system, and repository.

Pi is mainly the second part, with a useful default tool set and a terminal interface. Its lower-level packages can also be embedded in other applications. Pi is not an inference server (a program that hosts a model and answers requests) and does not make an ordinary model browser-capable. If the model has no search capability and Pi has no search tool, the agent has no dependable web search. It can run `curl` through the `bash` tool, but that is an execution primitive, not a research system.

A request follows this shape:

```
your instruction
  -> session and system prompt
  -> provider and model
  -> model response or tool call
  -> Pi executes the tool
  -> result enters the session
  -> model continues
```

The model decides what it wants to do, Pi executes the request, and the operating system decides what the process may actually do.

## 2.2 What Pi Provides

Pi supplies a terminal UI, provider and model selection, context-file discovery, persistent sessions, and a small default tool set. Pi enables four tools by default: `read`, `bash`, `edit`, and `write`. The `grep`, `find`, and `ls` tools are built in but disabled unless you allow them with `--tools` or the `defaultTools` setting.

That is enough for:
`inspect repository -> edit file -> run tests -> inspect failure -> edit again`

Pi includes an MCP client as a built-in extension, while planning, subagents, browser automation, and general web search remain workflow choices. Optional `codemode` lets the model compose tool calls in JavaScript; it is off by default unless selected or enabled by an MCP server. See the CLI tool reference and codemode reference. You can install a package, write an extension, create a skill, or keep other capabilities outside Pi and feed their results into the session.

Pi runs interactively, prints one-shot output, emits JSON events, communicates over remote procedure call (RPC), or is embedded through its TypeScript software development kit (SDK), so it is both a terminal program and a component of larger applications.

## 2.3 Why Pi Is Built This Way

The design is an argument, not an accident. The author's essay What I learned building an opinionated and minimal coding agent (November 2025) made it in five moves. Read these as the original design rationale, not a current feature list:

- **A tiny prompt.** The system prompt plus tool definitions come in below 1,000 tokens, and, the author argues, four tools suffice for coding work.
- **No MCP in the original core design.** The Model Context Protocol (MCP) is the standard for plugging such servers into an agent. The essay measured Playwright MCP at 21 tools and 13.7k tokens, and Chrome DevTools MCP at 26 tools and 18k tokens, "7-9% of your context window gone before you even start working". Pi 1.0 now ships a built-in MCP client with selective tool exposure; the essay remains useful as historical design rationale, not as a current feature list.
- **No built-in subagents, plan mode, or to-dos.** Subagents hide what the delegate did; plans and task lists belong in files you can read and edit.
- **No background bash.** The `bash` tool runs synchronously; use `tmux` for long-running processes.
- **YOLO by default.** Pi asks no permission before file operations or commands, and expects you to contain it externally.

The essay also reports a complete Terminal-Bench run as evidence. Treat that as the author's argument, not a neutral benchmark; the trade-off is that you supply much of the workflow yourself. The MCP position has since changed: current built-in MCP support uses selective tool exposure rather than requiring every server schema in every prompt. On licensing, the author states that Pi's core "will stay MIT licensed", while some future commercial features may be Fair Source or proprietary.

### 2.4 Coming From Claude Code

An existing Claude Code repository half-works on first run: Pi loads `CLAUDE.md` as a context file and implements the same Agent Skills format that Claude Code skills use, but it does not scan `.claude/skills`, so skills must be copied or pointed at. Most other concepts have a Pi counterpart, though often as an example or package rather than a built-in.

| Claude Code | Pi | Where it lives |
|---|---|---|
| `CLAUDE.md` | `AGENTS.md` or `CLAUDE.md`, both loaded from `~/.pi/agent`, the working directory, and its parents; `-nc` (`--no-context-files`) disables them | Core |
| Skills (`SKILL.md`) | Same Agent Skills format. Pi reads `~/.pi/agent/skills/`, `.pi/skills/`, `~/.agents/skills/`, and `.agents/skills/`, not `~/.claude/skills` | Core |
| Custom slash commands | Prompt templates in `prompts/` | Core |
| Hooks (`PreToolUse` and others) | Extension events such as `pi.on("tool_call", ...)`, which can change or block a call | Core, in TypeScript |
| Plugins | Pi packages, installed with `pi install` | Core |
| `claude -p` | `pi -p` (`--print`) | Core |
| `--continue`, `--resume`, `/compact` | Same names | Core |
| Rewind and branching (`/rewind`, `/branch`) | `/tree`, `/fork`, `/clone` branch the conversation; none of them rolls back file changes (the `git-checkpoint.ts` example does) | Core |
| `settings.json` | `~/.pi/agent/settings.json` and `.pi/settings.json` | Core |

| Claude Code | Pi | Where it lives |
|---|---|---|
| Permission prompts and modes | None; the `permission-gate.ts` example extension, or a container | Example extension |
| Plan mode | None; the `plan-mode/` example extension, or write plans to files | Example extension |
| To-dos | None; the `todo.ts` example extension | Example extension |
| Subagents | None; the `subagent/` example, the `pi-subagents` package, or more Pi processes in `tmux` | Example or package |
| MCP servers | Built-in stdio and Streamable HTTP client; `mcp.json`, `pi mcp`, and `/mcp` | Built-in extension |
| Tool-call composition | Optional `codemode` JavaScript tool; selective discovery through `tool_search` | Built-in extensions |
| `WebFetch` and `WebSearch` | None; add a skill or package | Not provided |

Pi keeps credentials, trust decisions, and sessions under `~/.pi/agent/`, not `~/.claude`. Because it does not ask before each tool call, the safety habits you built on Claude Code's prompts do not transfer; see "Safety and containment".

## 3 Installation and First Run

Pi is a Node.js package. The npm route needs Node.js 22.19 or newer; the installer checks for Node and offers to install it. The quickstart distinguishes the managed installer, which pins dependencies, from npm, which no longer pins transitive dependencies. The two routes are alternatives, not a sequence:

```
# Option A: the installer (macOS and Linux). Download and read it before running.
curl -fsSL https://pi.dev/install.sh -o pi-install.sh
grep -n 'npm install\|curl' pi-install.sh   # ~1,700 lines; find what it runs
sh pi-install.sh

# Option B: npm (Node.js 22.19 or newer)
npm install -g --ignore-scripts @earendil-works/pi-coding-agent

# Either way:
pi --version
pi --help
```

Older tutorials use `@mariozechner/pi-coding-agent` or the `badlogic/pi-mono` repository. That npm package is deprecated and frozen at 0.73.1; new releases exist only as

`@earendil-works/pi-coding-agent`, and the repository now lives at `earendil-works/pi`. Some companion repositories, including `badlogic/pi-skills`, still sit under the original account.

`pi update` updates a managed Pi installation (`pi update --extensions` updates packages). In 1.0.1 and later, a global npm installation recommends migrating to the managed installer; follow your installation method's update instructions. Pi 1.0 also defaults to a fullscreen terminal UI; `pi --tui-mode regular` or `tuiMode: "regular"` preserves normal terminal scrollback. These changes are documented in the changelog. To remove an npm install, run `npm uninstall -g @earendil-works/pi-coding-agent`; that leaves `~/.pi/agent/` (credentials, sessions, settings) in place.

### 3.1 A Controlled First Session

The first session should prove the complete control loop, not demonstrate how much work Pi can do unattended. Use a disposable project:

```
mkdir -p ~/tmp/pi-first-run && cd ~/tmp/pi-first-run
git init
printf '# Pi test project\n' > README.md
pi --version
```

Then work through this sequence:

1. **Authenticate and choose one model.** Use `/login` or one environment variable (see "Hosted authentication" in the next section), then `/model`. Start with one tool-capable hosted model or one already-tested local model. Do not add routing or fallbacks before the basic loop works.
2. **Orient with restricted tools.** Start Pi with `pi --no-mcp --tools read,grep,find,ls`, then send the prompt below. This removes editing and shell tools and disables built-in MCP for the run; `--tools` alone now keeps MCP tools unless it explicitly filters them. Loaded extensions still execute with your permissions, so this is a tool-selection check, not operating-system containment.
3. **Check what Pi loaded.** The working directory is the project: Pi discovers instructions and configuration from it and groups sessions by it. The startup header lists the context files and skills that loaded. If the folder holds `.pi/` resources or `.agents/skills`, Pi first asks whether to trust the project (see "Safety and containment").
4. **Make one supervised change.** Restart without `--tools`. Choose a small documentation or test change. Ask Pi to explain its plan, edit only the named files, run one check, and stop.

5. **Inspect independently.** Run `git status`, `git diff`, and the project's check command yourself. Keep or revert the change on evidence, not on Pi's confidence.

```
Inspect this repository. Report the files, likely project type,
and three useful next checks.
```

### 3.2 Working in the Terminal

You can send input while Pi works. `Enter` sends a steering message, which guides the next response once the current tool calls finish; `Alt+Enter` queues a follow-up that waits until Pi has finished all pending work; `Escape` aborts. `@` searches for a file to add to the prompt, `!git status` runs a command and sends its output to the model, `!!git status` runs it without sending the output, `/model` and `/thinking` change the model and its reasoning level, and `Shift+Enter` adds a line break.

## 4 Models and Authentication

Pi reaches a model through a provider. This section covers signing in to a hosted provider first, then compatible and local models.

### 4.1 Hosted Authentication

Either export an API key, or sign in from inside Pi. These are alternatives:

```
export ANTHROPIC_API_KEY='<your-api-key>'    # environment variable: suits ephemeral runs
pi
/login
```

`/login` stores an API key or, for some providers, a subscription credential in `~/.pi/agent/auth.json`. Protect that file, and keep secrets out of settings files, prompts, skills, and committed scripts. When several sources exist, Pi uses a `--api-key` flag first, then `auth.json`, then a `models.json apiKey`, then environment variables, so a stale stored login beats a freshly exported variable. `pi auth check --provider <name>` reports `ready`, `not_ready`, or `invalid` without printing a secret. Run `/model` afterwards to select a model.

**Claude subscriptions need a caution.** Claude Code readers will reach for the subscription they already pay for. What is confirmed: when Anthropic subscription authentication is active, Pi warns that third-party harness usage draws from "extra usage" billed per token, not from your plan limits, and added a `warnings.anthropicExtraUsage` setting to silence the warning (issue #3808, April 2026, closed as completed). What is not settled: Anthropic's support page on Agent SDK usage says a planned change was paused on 15 June 2026 and that Agent SDK, `claude -p`, and third-party app usage still draw from subscription limits, while a contributor in the same issue thread reports that OAuth use by harnesses such as Pi is still billed as extra usage. This primer found no Anthropic page that resolves the conflict.

An API key is the unambiguous route; check Anthropic's current terms before signing in with a subscription.

Pi's built-in catalogue covers more than 15 providers, including Anthropic, OpenAI, Google Gemini, Azure OpenAI, Vertex, Bedrock, OpenRouter, Mistral, Groq, xAI, DeepSeek, and GitHub Copilot. It is bundled and refreshed from pi.dev (`pi update --models` forces a refresh), so use `/model` or the model catalogue for current IDs and prices rather than copying an old model ID.

## 4.2 Hosted and Local Models

Provider flexibility is not the same as model interchangeability. Tool calling, streaming, context length, authentication, image support, and reasoning controls vary. A model that answers chat questions may still be a poor coding-agent model if it produces unreliable tool calls.

### 4.2.1 OpenAI-Compatible Local Servers

Ollama, LM Studio, vLLM, and SGLang (inference servers) can expose OpenAI-compatible endpoints. Pi reads them from `~/.pi/agent/models.json`, with no code:

```
{
  "providers": {
    "ollama": {
      "baseUrl": "http://127.0.0.1:11434/v1",
      "api": "openai-completions",
      "apiKey": "ollama",
      "models": [{ "id": "<installed-model-id>" }]
    }
  }
}
```

The `apiKey` is a dummy: Pi needs a key to treat the model as available, and Ollama ignores it. Replace the placeholder with the exact model ID served by your installation, then test its tool calling; a small model's ability to answer chat questions is insufficient. Check the context window the server allocates before blaming Pi for a lost instruction, because the server sets it. The models documentation is current.

A useful arrangement is a local model for routine exploration, with a hosted model as an explicit, per-task fallback for code you are willing to send to that provider. A local model does not make all data local: a search extension, browser, provider gateway, or external API can still receive content. Pi itself sends anonymous install and update telemetry by default (see "Safety and containment").

### 4.2.2 llama.cpp

Pi has a dedicated integration with llama.cpp's router server, which discovers several GGUF files (llama.cpp's model file format) and loads them on demand. Start `llama-server` without `--model`, `-m`, or `-hf`; passing a model starts single-model mode, and Pi's router integration then fails. It needs a llama.cpp build with router support.

```
llama-server \
  --models-dir ~/models \
  --no-models-autoload \
  --jinja \
  --host 127.0.0.1 \
  --port 8080 \
  -ngl 999 \
  -c 32768
```

`--models-dir` is where the GGUF files live, `--no-models-autoload` keeps loading explicit, `-ngl 999` offloads all layers to the GPU, and `-c 32768` sets the context per loaded model (omit it for the model's native context, which may need much more memory). The `--jinja` flag enables the model's chat template and tool calling; without it, tool calls are unreliable.

In Pi, run `/login llama.cpp` (default URL `http://127.0.0.1:8080`), then `/llama` to load a model, then `/model` to select it. `LLAMA_BASE_URL` and `LLAMA_API_KEY` configure the same connection without `/login`. Check the server with `curl http://127.0.0.1:8080/health` and `curl http://127.0.0.1:8080/models`.

Test the layers in order: health check, ordinary chat, streaming, Pi-compatible tool calls, and finally Pi executing a tool and feeding the result back. Keep one known-good hosted fallback while you stabilise the local path.

# 5 Running Pi Non-Interactively

The table compares Pi's main execution modes and the work each suits.

| Way to run Pi | Interface | Use it when |
|---|---|---|
| Interactive | Terminal UI | A person is working with Pi |
| Print (`--print`, `-p`) | Final assistant text on stdout, then exit | A script needs the answer |
| JSON (`--mode json`) | JSON Lines (one JSON event per line) on stdout, then exit | A program needs structured progress |

| Way to run Pi | Interface | Use it when |
|---|---|---|
| RPC (`--mode rpc`) | JSON Lines commands on stdin, responses and events on stdout, long-lived | Another program drives Pi (RPC is remote procedure call: commands and replies over a pipe) |
| SDK | TypeScript library inside your Node.js or Bun process | You embed Pi in an application |

```
pi --print 'Summarize this repository without editing it'
git diff --no-ext-diff | pi --print 'Review this diff. Return findings only.'
pi --no-session --print 'Explain this directory'     # in-memory session, nothing saved
pi --mode json 'Inspect the repository and report its test commands' > pi-events.jsonl
pi --mode rpc --no-session   # driven by another program; not interactive
```

When stdin or stdout is redirected and no mode is chosen, Pi uses print mode, and piped stdin is prepended to the prompt. For read-only review, restrict the tools:

```
pi --no-mcp --tools read,grep,find,ls --print 'Review without changing files'
```

`--tools` selects built-in, extension, and custom tools, but retains MCP unless an entry begins with `mcp__`. Use `--no-mcp` to disable the built-in MCP client for this review; a replacement extension needs its own controls. The CLI reference lists every flag. To embed Pi in TypeScript, the SDK creates a session in your own process; RPC is the language-independent alternative.

```
import { createAgentSession } from "@earendil-works/pi-coding-agent";

const { session } = await createAgentSession();
try {
  await session.prompt("What files are in the current directory?");
  console.log(session.getLastAssistantText());
} finally {
  session.dispose();
}
```

# 6 Customisation: Use the Smallest Layer That Works

Pi's customisation layers run from plain instructions to executable code, and each step up adds power and review burden. The subsections below go through the layers in that order, then turn the choice into a table and a starter set.

## 6.1 Context Files and Prompt Templates

`AGENTS.md` and `CLAUDE.md` suit durable repository guidance: test commands, style, directory boundaries, and review requirements. Pi loads either from `~/.pi/agent`, the working directory, and its parents. `AGENTS.override.md` replaces them in the same directory only. `SYSTEM.md` replaces Pi's default system prompt and `APPEND_SYSTEM.md` extends it, at user or project level. Keep credentials and volatile provider details out of all of them.

Prompt templates play the role of Claude Code's custom slash commands. A Markdown file in `~/.pi/agent/prompts/` or `.pi/prompts/` becomes a command named after the file, and `$1`, `$@`, and `${1:-default}` substitute arguments. This one, saved as `review.md`, gives you `/review` and `/review concurrency`:

```
---
description: Review staged git changes
argument-hint: "[focus]"
---
Review the staged changes. Focus on ${1:-correctness, security, and error handling}.
```

Run `/reload` after editing a template in an active session.

## 6.2 Skills

A skill is on-demand instructions with optional scripts, references, and assets. Pi implements the open Agent Skills specification, the format Claude Code skills use, so a Claude Code skill directory generally works unchanged. Pi does not scan `~/.claude/skills`: copy the directory into `~/.pi/agent/skills/` or `~/.agents/skills/`, or point Pi at it with `--skill <path>` or the `skills` setting. Only each skill's name, description, and path sit in the prompt; the model reads the full `SKILL.md` when a task matches, or when you force it with `/skill:name`. Setting `disable-model-invocation: true` restricts a skill to that explicit command.

A project skill is a directory such as `.pi/skills/web-research/` holding a `SKILL.md`, plus optional `scripts/` and `references/`. The `SKILL.md` stays a short procedure:

```
---
name: web-research
description: Research a question from dated primary sources. Use for current docs and releases.
---

# Web research

Follow references/source-policy.md. Run scripts/search.sh "<query>".
```

If a skill needs a model-callable tool, it is becoming an extension. Project skills load only after you trust the project.

## 6.3 Extensions

Extensions are TypeScript modules executed inside Pi, which runs them directly with no build step. They can register tools, slash commands, lifecycle handlers, custom UI, provider adapters, and context or compaction behaviour. For a Claude Code user, one mechanism covers hooks, custom tools, and plugin code.

```
pi --extension ./my-extension.ts    # or -e; repeatable
~/.pi/agent/extensions/             # global
.pi/extensions/                     # project-local; loads only after you trust the project
```

A subdirectory containing `index.ts` also loads, and `/reload` picks up changes. The `project_trust` event can be handled only by personal and command-line extensions.

A custom tool has a name, a parameter schema, and an `execute` function. This one counts the words in a string, adapted from Pi's `hello.ts` example:

```
import { Type } from "@earendil-works/pi-ai";
import { defineTool, type ExtensionAPI } from "@earendil-works/pi-coding-agent";

const countWords = defineTool({
  name: "count_words",
  label: "Count words",
  description: "Count the words in a piece of text.",
  parameters: Type.Object({
    text: Type.String({ description: "The text to count" }),
  }),

  async execute(_toolCallId, params, _signal, _onUpdate, _ctx) {
    const n = params.text.split(/\s+/).filter(Boolean).length;
    return {
      content: [{ type: "text", text: `${n} words` }],
      details: { words: n },
    };
  },
});

export default function (pi: ExtensionAPI) {
  pi.registerTool(countWords);
}
```

Throwing from `execute` is how a tool reports failure; returning an object does not mark an error. A tool that touches the filesystem must validate its inputs, because it lets the model reach anything the user can.

The counterpart to Claude Code's permission prompts is a `tool_call` handler, which can change a call's input or block it. This gate, trimmed from the shipped `permission-gate.ts`, asks before a dangerous shell command and blocks it when there is no UI to ask (print and JSON modes; an RPC client can answer the dialog):

```
import type { ExtensionAPI } from "@earendil-works/pi-coding-agent";

export default function (pi: ExtensionAPI) {
  pi.on("tool_call", async (event, ctx) => {
    if (event.toolName !== "bash") return undefined;
    const command = event.input.command as string;
    if (/\brm\s+(-rf?|--recursive)|\bsudo\b/i.test(command)) {
      if (!ctx.hasUI) return { block: true, reason: "Dangerous command blocked" };
      const choice = await ctx.ui.select(`Allow this command?\n\n  ${command}`, ["Yes", "No"]);
      if (choice !== "Yes") return { block: true, reason: "Blocked by user" };
    }
    return undefined;
  });
}
```

Start from the shipped examples rather than from scratch: `permission-gate.ts`, `protected-paths.ts` (blocks writes to `.env`, `.git/`, and `node_modules/`), `confirm-destructive.ts`, `plan-mode/`, `subagent/`, `todo.ts`, `handoff.ts`, and `claude-rules.ts`. From a checkout of the repository, `pi -e packages/coding-agent/examples/extensions/permission-gate.ts`

gives you the gate above. These are prompts inside the Pi process, not a security boundary: a regular expression is easy to evade, and the extension runs with your permissions. An extension can read credentials, alter prompts, intercept tool calls, and start child processes. Review it like application code.

### 6.4 Packages

A package bundles extensions, skills, prompt templates, and themes (JSON colour files) for distribution through npm or Git; it is the counterpart of a Claude Code plugin. The package catalogue lists packages that carry the `pi-package` npm keyword. It is for discovery, not a security review.

```
pi install npm:@scope/package@1.2.3      # scoped npm package, version-pinned
pi install git:github.com/vendor/repo@<reviewed-commit>
pi install -l npm:@scope/package@1.2.3   # project scope; read only after trust
pi -e npm:@scope/package                  # try for one run, without installing
pi list
pi config                                 # enable or disable resources
pi remove npm:@scope/package@1.2.3
```

Versioned npm specifications, Git tags, and commits are pinned: `pi update --extensions` reconciles installations but does not move a configured ref. Read the source, installation scripts, and network calls before installing, and test third-party packages in a disposable project. A package declares its resources through conventional `extensions/`, `skills/`, `prompts/`, and `themes/` directories or a `pi` key in `package.json`.

### 6.5 MCP and Subagents

Pi now has built-in MCP support over stdio and Streamable HTTP. User servers live in `~/.pi/agent/mcp.json`; project servers live in `.pi/mcp.json` and require project trust. Inspect them with `pi mcp list` or `/mcp`. Default `codemode` exposure lets scripts discover and call tools without declaring every schema; `direct`, `deferred`, and `hidden` exposure serve different needs. OAuth and bearer authentication are documented there. Start with one read-only server and verify its credentials, scope, and outputs before adding write tools.

The community `pi-mcp-adapter` is now an alternative implementation, not a prerequisite; avoid loading competing clients accidentally. Subagents remain an example or package workflow: use `subagent/`, `pi-subagents`, or further Pi processes in `tmux` when needed.

### 6.6 The Starter Suite: Install by Need

There is no universally correct add-on bundle. Pi's quickstart says to start with the least powerful mechanism that meets your need, and its security page notes that extensions run inside the Pi process with your permissions. Every executable extension adds code to review

and another source of prompt injection (instructions hidden in files, web pages, or tool output that the model may obey) or data leakage.

| Need | Start with | Move to a heavier mechanism when |
|---|---|---|
| Repository conventions | `AGENTS.md` (or your existing `CLAUDE.md`) | You repeat a procedure across repositories |
| Reusable procedure | A project or user skill | The procedure needs a model-callable tool or lifecycle hook |
| Repeated review prompt | A prompt template | The workflow needs state, events, or custom UI |
| Public web research | `brave-search` from `pi-skills` | You need another provider or richer extraction |
| JavaScript-heavy or authenticated pages | `browser-tools` from `pi-skills` | You need screenshots or frontend QA (see the caveats in "Add a browser only when needed") |
| New model or endpoint | A `models.json` entry, or `/login llama.cpp` | The service needs a custom protocol or authentication flow |
| Approval gate or new tool | A reviewed extension | You want to share it: publish a package |

Instructions are easier to audit than executable customisation. A worked path: to have Pi review pull requests, start with a paragraph in `AGENTS.md`, then a `/review` prompt template, then a skill if the procedure grows scripts, and an extension only if you need a gate or a tool.

The `pi-skills` repository, kept by Pi's original author, is the sensible first add-on for research. Clone it somewhere neutral, pin what you reviewed, and load only the skill you need. Pi scans skill locations recursively and puts every discovered skill's name and description in the system prompt, so cloning the whole repository into a skills directory advertises its Gmail, Drive, and Calendar clients to the model.

```
git clone https://github.com/badlogic/pi-skills ~/src/pi-skills
git -C ~/src/pi-skills checkout <reviewed-commit>          # pin what you reviewed
(cd ~/src/pi-skills/brave-search && npm install --ignore-scripts)
pi --skill ~/src/pi-skills/brave-search                    # load it for this run only
```

To load it in every session, copy the skill directory into `~/.pi/agent/skills/` or add its path to the `skills` setting.

`brave-search` also needs a Brave Search API key: create an account at api-dashboard.search.brave.com, add a "Free AI" subscription (the skill notes that a credit card is required even for the free plan), and `export BRAVE_API_KEY=...` in the shell that launches Pi. The skill sends your search queries to Brave.

The community package `pi-web-access` (by Nico Bailon, MIT; `pi install npm:pi-web-access@0.31.0`) combines search, fetching, and extraction. Treat it as a candidate to inspect, not a default. With no configuration it searches through Exa's hosted MCP endpoint, so queries reach a third party without you adding a key. Its optional answer and summary modes call a separate model and can send fetched page text to a different provider. Browser-cookie access is opt-in (`allowBrowserCookies: true` or `PI_ALLOW_BROWSER_COOKIES=1`), and settings live in `~/.pi/agent/web-search.json`.

Start with the built-in MCP client if you need MCP; do not install a second adapter by habit. Add a subagent framework, memory system, planning overlay, or larger bundle only when you can name the recurring problem it solves and the boundary it requires.

# 7 Web Research: Assemble the Missing Capability

Pi ships no general browser or web-search tool. A task that needs current documentation can use `curl` through `bash`, a skill, or a configured MCP search service; the MCP client does not itself supply a search backend.

## 7.1 Plain HTTP and Git Are Good for Static Sources

```
curl -fsSL -o extensions.md \
  https://raw.githubusercontent.com/earendil-works/pi/v1.0.4/packages/coding-agent/docs/extensions.md
curl -fsSL https://api.github.com/repos/earendil-works/pi/releases/latest
git clone --depth 1 --branch v1.0.4 https://github.com/earendil-works/pi.git ~/src/pi
```

This suits raw Markdown, JSON APIs, release metadata, and source trees. Pin a tag or commit rather than `main` when you need the same bytes next week.

## 7.2 Plain HTTP Is Not Web Research Infrastructure

A plain HTTP request does not search, render JavaScript, keep signed-in sessions, click through consent banners, or extract readable text from a complex page. A successful status can still return only an application shell: the near-empty HTML page that a JavaScript single-page app fills in later. Do not ask Pi to "search the web" and accept model-memory guesses; give it an explicit tool, source policy, evidence format, and stop condition.

### 7.3 Start With a Maintained Search Skill

The pi-skills repository, kept by Pi's original author under his own account (last pushed June 2026: maintained, not fast-moving), holds eight skills: Brave web search, Chrome DevTools browser automation, Google Calendar, Drive, and Gmail clients, Groq speech-to-text, VS Code diffs, and YouTube transcripts. They also work in Claude Code, Codex CLI, Amp, and Droid. Installation, pinning, and the Brave key are under "The starter suite", in the Customisation section.

`/skill:name` forces a skill to load, which matters because a model can fail to load a relevant skill on its own; text after the command is appended as your request (the same pattern as Claude Code's `/name`). Use `Shift+Enter` for line breaks so the text is one message:

```
/skill:brave-search

Research the current Pi extension API. Search official documentation first,
then the repository, issues, discussions, and release notes. For each important
claim record URL, date, source type, and confidence. Do not install packages or
modify files. Return a source table before the synthesis.
```

These are the helper scripts the skill tells the model to run; run them yourself to test your key:

```
~/src/pi-skills/brave-search/search.js \
  'Pi extension registerTool' -n 10 --content --freshness pm

~/src/pi-skills/brave-search/content.js \
  https://raw.githubusercontent.com/earendil-works/pi/v1.0.4/packages/coding-agent/docs/extensions.md
```

`-n` sets the result count (default 5, maximum 20), `--content` fetches each page as Markdown, `--freshness` filters by `pd`, `pw`, `pm`, `py`, or a `YYYY-MM-DDtoYYYY-MM-DD` range, and `--country` sets the region (default US). Prefer a raw Markdown URL to a GitHub `blob` page, whose HTML adds navigation noise. Keep the API key in the environment, not in the skill.

Earendil also runs Radius, a hosted, credit-metered gateway for Pi whose documentation lists web search among its tools: a first-party alternative to assembling search yourself, with the data-path questions any hosted service raises.

### 7.4 Add a Browser Only When Needed

For JavaScript-heavy documentation, authenticated forums, or frontend verification, `browser-tools` drives Chrome through the Chrome DevTools Protocol on port 9222 and extracts content after page load. Four caveats:

- **It is macOS-only as shipped.** `browser-start.js` hard-codes the macOS Chrome path (issue #33; a cross-platform pull request #47 is unmerged as of 24 September 2026). On Linux, patch it, launch Chrome yourself with

`--remote-debugging-port=9222 --user-data-dir=<throwaway dir>`, or use a different tool.

- **The default profile is persistent, not disposable.** Without flags it uses `~/.cache/browser-tools`; delete it between runs. `--profile` copies your real Chrome profile (cookies and logins) into it; do not use that flag with an agent.
- **Anything that can reach port 9222 can drive the browser.** The skill can dump cookies (`browser-cookies.js`) and run arbitrary JavaScript in a page (`browser-eval.js`) into the model's context. Treat the browser, the model, and every extension as one trust domain, and use a narrow account.
- **It uses bot-detection evasion** (`puppeteer-extra-plugin-stealth`), which matters for site terms of service on authenticated forums.

Use this decision rule:

| Need | Appropriate first tool |
|---|---|
| Static documentation, JSON, raw GitHub | `curl` or a narrow shell skill |
| Search-engine results | Search API skill |
| JavaScript-rendered public page | Browser automation, or a hosted readability converter |
| Authenticated forum | Disposable browser profile and explicit scope |
| Frontend testing | Browser automation with screenshots and logs |

### 7.5 Search GitHub and Forums Directly

`gh` is the GitHub command-line client; run `gh auth login` first, because code search needs authentication.

```
gh search code registerTool --repo earendil-works/pi
gh search issues 'MCP' --repo earendil-works/pi
gh api repos/earendil-works/pi/discussions \
  --jq '.[] | "\(.number)\t\(.title)\t\(.html_url)"'
```

Search results are leads, not evidence: read the issue, check its date and status, and distinguish a maintainer statement from a user workaround. A useful research brief states the decision it must support, the source order (official docs and source, release notes, issues, then community material), the required freshness, the evidence format (URL, date, claim, confidence), and a stop condition such as ten sources or three independent confirmations. For example:

```
Determine whether Pi currently has a built-in MCP client.
Search the official repository and docs first, then issues and the package
catalogue. Do not infer core support from an extension. Return URL, date,
```

```
source type, exact claim, and confidence. Stop after the current README, docs
index, source tree, and five relevant issues or package entries.
```

The answer as of 1.0.4 is yes: the official MCP documentation describes the built-in client, configuration, transports, and tool exposure. Older material correctly described earlier releases without it. This brief guards against treating an old design essay or forum answer as current behaviour, and against confusing a community adapter with the shipped client.

# 8 Sessions, Hand-Off and Workflow

This section covers how Pi stores and branches conversations, how to carry work from one session to the next, and a five-step routine that fits sessions into a task.

## 8.1 Sessions and Compaction

Pi sessions are persistent conversation records stored as JSON Lines files under `~/.pi/agent/sessions/`, grouped by working directory. Override the location with `--session-dir`, `PI_CODING_AGENT_SESSION_DIR`, or the `sessionDir` setting. The active conversation is one branch of a tree in that file.

```
pi --continue        # reopen the latest session for this directory (as in Claude Code)
pi --resume          # pick a saved session
pi --name research   # name a session; /name does the same from inside
pi --no-session      # in-memory only
```

| Command | Result | Use it when |
|---|---|---|
| `/tree` | Moves within the current session file | Related alternatives should stay together |
| `/fork` | New session from an earlier user message | The alternative should become separate work |
| `/clone` | Copies the active branch into a new session | You want a separate copy of the current state |

`/session` shows the file, ID, token usage, and cost; `/import` and `/export` move sessions in and out.

Pi compacts automatically as the context fills: with the defaults, when it comes within 16,384 tokens of the model's limit, keeping roughly the most recent 20,000 tokens. Compaction adds a summary and keeps recent messages; it does not delete the original entries, which stay in the session file. What degrades is what the model sees on later requests, so check what the summary dropped before relying on an old detail. Run `/compact <instructions>` to steer what is kept.

### 8.2 Sharing and Hand-Off

Sessions may contain source code, prompts, command output, URLs, tool arguments, and accidental secrets. Treat them as sensitive data. `/share` uploads the whole session and returns a viewer link (a private GitHub gist unless you use Radius), and `/bug` can attach the transcript, so review both first. For a durable hand-off, write:

```
objective
files changed
commands run and results
open questions
assumptions
next safe step
```

The shipped `handoff.ts` example extension does this in one step: `/handoff <next task>` creates a new session with a generated prompt for you to review.

### 8.3 A Practical Workflow

1. **Map the repository.** Use a read-only prompt to identify instructions, build commands, tests, generated files, and the smallest safe change.
2. **Research in a separate session.** Use a search skill or API. Require dated sources and separate facts from inferences.
3. **Hand off evidence.** Write a short reviewed research note with URLs, dates, excerpts, and unresolved uncertainty.
4. **Implement in a branch or worktree.** Give Pi one bounded change, explicit file ownership, and proof commands.
5. **Verify independently.** Run formatter, tests, static checks, and rendering outside the model's narrative. Review the diff and working-tree state.

Name each session (`--name` or `/name`) so hand-off notes can refer to it. This keeps research context from overwhelming implementation context and makes the result auditable, if not reproducible.

## 9 Safety and Containment

Pi is not a sandbox and has no built-in permission boundary for filesystem, process, network, or credential access. It runs with the launching user's permissions. Bash, extensions, package installers, language servers, and child processes can affect whatever that user can affect. Claude Code, by contrast, offers per-tool permission prompts and an optional OS-level sandbox for shell commands (Seatbelt on macOS, bubblewrap on Linux); Pi has neither built in, and its security documentation says to use one of the isolation patterns below.

Three controls are easy to confuse:

- Project trust controls whether project-local resources load.
- Tool selection and MCP exposure control which tools are callable; `--tools` alone does not remove MCP tools.
- The operating system, container, VM, or sandbox controls what the process can touch.

## 9.1 Project Trust

Trust exists to stop a folder silently loading executable extensions or project MCP servers. Pi asks for a decision when the working directory holds any of these: `.pi/settings.json` or `.pi/mcp.json`; `.pi/extensions`, `.pi/skills`, `.pi/prompts`, or `.pi/themes`; `.pi/SYSTEM.md` or `.pi/APPEND_SYSTEM.md`; or `.agents/skills` in the working directory or an ancestor. A bare `.pi` directory needs no decision. Declining skips those resources. The security guide describes the complete loading boundary.

Context files are not gated. `AGENTS.md`, `CLAUDE.md`, and `AGENTS.override.md` load regardless of trust unless you pass `-nc`, so a hostile repository can still steer the model through them even if you decline trust. One documented gap: the project `sessionDir` setting is read before trust is resolved.

You decide interactively, save a decision with `/trust` (stored in `~/.pi/agent/trust.json`), or override for one process with `--approve` (`-a`) or `--no-approve` (`-na`). The `defaultProjectTrust` setting is `ask` (the default), `always`, or `never`. Print, JSON, and RPC modes cannot prompt, so under `ask` they skip protected resources unless a saved decision or `--approve` applies: a CI script that expects a `.pi/` extension to load silently will not load it. Trust is not containment: after startup, enabled tools still use the process's operating-system permissions.

## 9.2 Isolation Options

For untrusted repositories, unattended runs, production credentials, or browser profiles, use a container or VM. A container is only as strong as its mounts, environment variables, network access, and user identity. Pi's containerisation guide documents four methods:

| Method | Where Pi runs | What is isolated | Credentials |
|---|---|---|---|
| Plain Docker | Container | Pi, its tools, `!` commands, extensions | Passed into the container |

| Method | Where Pi runs | What is isolated | Credentials |
|---|---|---|---|
| Docker Sandboxes (`sbx`) | Managed sandbox | The same | Stay on the host; a proxy substitutes them |
| NVIDIA OpenShell | Local or remote sandbox | The same | Policy-controlled |
| Gondolin extension | Host | Built-in tools and `!` commands only, in a local micro-VM | Commands inherit host environment variables, so it is not a credential boundary |

Plain Docker is the simplest. Pi's documentation builds the image from this `Dockerfile.pi`:

```
FROM node:24-bookworm-slim
RUN apt-get update \
  && apt-get install -y --no-install-recommends bash ca-certificates git ripgrep \
  && rm -rf /var/lib/apt/lists/*
RUN npm install -g --ignore-scripts @earendil-works/pi-coding-agent
WORKDIR /workspace
ENTRYPOINT ["pi"]
```

```
docker build -t pi-sandbox -f Dockerfile.pi .
docker run --rm -it \
  -e ANTHROPIC_API_KEY \
  -v "$PWD:/workspace" \
  -v pi-agent-home:/root/.pi/agent \
  pi-sandbox
```

`podman run` works the same way; on an SELinux host add `:Z` to the bind mount. Passing `-e ANTHROPIC_API_KEY` puts the key inside the container, where every process, extension, and tool call can read it. The named `pi-agent-home` volume keeps settings, credentials, and sessions between runs; mounting the host's `~/.pi/agent` instead would expose your real credentials, extensions, and sessions. Do not mount the whole host home directory or a real browser profile, and give research extensions only the keys and network routes they require.

### 9.3 Prompt Injection, Telemetry, and Reporting

Prompt injection can come from a README, issue, web page, command output, or model response; treat external text as untrusted input. Pi's security policy treats prompt injection, the lack of a built-in sandbox, and user-installed extensions or skills as generally outside its security boundary; report genuine bypasses through `SECURITY.md`, not a public issue.

Pi also makes outbound requests of its own. Anonymous install and update reporting, plus some provider attribution headers, are on by default (`enableInstallTelemetry`); turn them

off with `PI_TELEMETRY=0`. The model catalogue refreshes from pi.dev. `pi --offline` (or `PI_OFFLINE=1`) disables automatic network activity, including that refresh.

## 10 Pi Compared With Other Coding Agents

The useful comparison is not a feature checklist. Ask where the product puts workflow decisions and what the default execution boundary is.

| Agent | Centre of gravity | Provider posture | Web posture | Customisation and controls |
|---|---|---|---|---|
| **Pi** | Programmable terminal harness and embeddable runtime (MIT) | Broad providers plus compatible and local endpoints | Add a skill, MCP service, extension, package, or shell workflow | TypeScript extensions, built-in MCP and optional codemode; no per-tool approval system |
| **OpenCode** | Ready-made terminal coding application | Broad provider and local support | Built-in `webfetch`; `websearch` only with the OpenCode provider or an opt-in Exa or Parallel flag | Per-tool permissions, plugins, skills, MCP |
| **Aider** | Git-aware pair programmer | Many providers and local endpoints | `/web <url>` scrapes a page into the chat; no built-in search | Configuration and scripts; narrower runtime |
| **Claude Code** | Opinionated terminal agent around Claude | Strong Anthropic alignment and supported cloud deployments | Built-in `WebFetch` and `WebSearch`, permission-gated | Permission modes, optional OS-level sandbox, subagents, hooks, skills, plugins, MCP |

| Agent | Centre of gravity | Provider posture | Web posture | Customisation and controls |
|---|---|---|---|---|
| **Codex** | Product-connected coding agent from OpenAI | OpenAI-first; custom providers and local `--oss` models supported | Built-in web search, cached results by default | Instructions, skills, plugins, MCP; sandbox modes and approval policies built in |

### 10.1 Pi Versus OpenCode and Aider

OpenCode feels like a complete coding application; Pi starts smaller and asks you to assemble the workflow. Choose OpenCode for a richer day-one experience. Choose Pi when the agent loop itself must be programmable, when you want a narrow, provider-neutral harness, or when you are building an embedded agent. Aider's centre is the repository, selected editable files, a repository map, and a reviewable Git edit loop. Pi can reproduce that loop but can also become a research tool, browser bridge, or application component; choose Aider for focused supervised edits.

### 10.2 Pi Versus Claude Code and Codex

Claude Code and Codex supply the surrounding workflow themselves: permission prompts, sandbox options, web search, and established instruction systems. The mapping table in "Coming from Claude Code" shows where Pi leaves each of those to you: you select the model, assemble research access, review extensions, and define containment. That flexibility is not automatically cheaper or safer. Local models add hardware and quality costs; extensions can add exactly the capability you want while gaining full process access. The sibling primers cover the alternatives to Claude Code and Codex for Claude Code users in more depth.

### 10.3 Forks and Derivatives

Because Pi is MIT-licensed and embeddable, others build on it. `oh-my-pi` (Stencil Labs, MIT) is, by its own README, a fork that adds subagents, integrated development environment (IDE) and Language Server Protocol (LSP) support, and browser tooling. OpenClaw (see the OpenClaw primer) was described in January 2026 as having Pi "under the hood" (Armin Ronacher); its current documentation says only `@earendil-works/pi-tui`, the terminal-UI toolkit, remains a dependency. A claim that a project "runs on Pi" needs a date.

# 11 Worked Example: A Reproducible Data-Science Project With Pi and Rocker

This project makes the boundary between an agent and an analysis runtime concrete. Pi runs on the host and helps inspect the repository, research package APIs, write code, and interpret failures. Rocker runs the R and Python analysis in a reproducible container, and Podman supplies the rootless boundary. Pi itself remains a host process with your permissions (see "Safety and containment"); the container bounds only the analysis code. Because Pi runs `just qa` on the host, it also has host access to Podman and everything else you have.

Terms: **Rocker** publishes versioned R container images (`rocker/r-ver`, `rocker/tidyverse`); the tidyverse image bundles R, `readr` and `dplyr` (data reading and manipulation), and `testthat` (R's unit-test framework), and is large (about 3.8 GB), where `rocker/r-ver` is much smaller. **Podman** is a daemonless container engine with a Docker-compatible command line; **rootless** means it runs without root privileges on the host. A **digest** is a content hash (`sha256:...`) identifying an exact image, where a tag is a movable label. `renv` is R's dependency lockfile tool.

## 11.1 Project Goal and Contract

Build a small "daily temperature quality" project. The input is a fictional CSV of station readings; two independent implementations, one in R and one in Python, must produce the same summary. Do not call Python from R (for example through the `reticulate` package): independence is what makes their agreement meaningful. You write this contract before Pi writes any code:

- Dates are ISO `YYYY-MM-DD`; a malformed date is an error. The allowed `quality_flag` values are `ok` and `missing`; any other value is an error, as is an `ok` row with an empty `temp_c`. An error means a non-zero exit status and a message naming the problem.
- Rows marked `missing` are discarded. A station-day with only `missing` rows produces no output row. Group the rest by `date` and `station_id`.
- The output header is exactly `date,station_id,n_valid,mean_temp_c,min_temp_c,max_temp_c`, sorted by `date`, then `station_id`, byte-wise (C-locale order; R's default string ordering depends on the locale). Format temperatures with two decimals; use UTF-8, LF line endings, and no quoting.
- Tolerance: keys, `n_valid`, `min_temp_c`, and `max_temp_c` must match exactly; `mean_temp_c` may differ by at most `0.01`, because R and Python can round a tie differently.

The repository layout:

```
pi-temperature-starter/
  AGENTS.md
  data/raw/station_readings.csv
  data/expected/daily_summary.csv
  r/clean.R
  r/testthat/test-clean.R
  python/clean.py
  python/tests/test_clean.py
  scripts/compare_outputs.py
  outputs/               # generated; git-ignored apart from .gitkeep
  renv.lock
  requirements.txt
  Containerfile
  .containerignore
  .gitignore
  Justfile
  README.md
```

`AGENTS.md` carries the rules Pi must follow, and Pi loads it automatically at startup:

```
# Project rules
- Run all analysis inside the container with `just qa`. Never run R or pip on the host.
- Do not edit `data/raw/`, `data/expected/`, `Justfile`, or `Containerfile`.
- Read the contract in `README.md`. Stop and ask if it is ambiguous.
```

Start with a deliberately small fixture. One group has two valid readings, so the mean, minimum, and maximum differ; the last row forms a group of only missing data:

```
date,station_id,temp_c,quality_flag
2026-01-01,A,5.1,ok
2026-01-01,A,6.3,ok
2026-01-01,A,,missing
2026-01-01,B,3.4,ok
2026-01-02,A,,missing
```

The expected output is part of the contract, not something Pi may invent after seeing its own result. Write it by hand:

```
date,station_id,n_valid,mean_temp_c,min_temp_c,max_temp_c
2026-01-01,A,2,5.70,5.10,6.30
2026-01-01,B,1,3.40,3.40,3.40
```

The fixture has no unknown flag or malformed date, so those failure cases live in the unit tests as small in-memory inputs. The acceptance gate is `scripts/compare_outputs.py`, which you write, not Pi. It reads `outputs/daily_summary_r.csv` and `outputs/daily_summary_py.csv`, checks each header, compares each row with the expected file (exact keys and counts, `mean_temp_c` within the tolerance), and exits non-zero on any difference. Row-by-row comparison against a sorted file also checks the sort order.

## 11.2 Give Pi Bounded Tasks

Use a separate, named session for each phase. The fixture, expected output, and comparison script are committed before the implementation task starts.

Phase 1 restricts tools: `pi --name contract --no-mcp --tools read,grep,find,ls`.

```
Inspect the repository rules and propose the file contract for this project.
Do not write files or install packages. Identify assumptions and test cases.
```

Phase 2 needs a search skill and `bash`: `pi --name research --skill ~/src/pi-skills/brave-search` (see "Web research"). Without a search skill, drop this step and supply the documentation URLs yourself; otherwise Pi answers from model memory.

```
Research the current official documentation for readr, dplyr, testthat,
Python csv handling, and pytest. Return URLs and version constraints only.
Do not modify the repository.
```

Phase 3 is implementation in a fresh session, `pi --name implement`, with the research hand-off note pasted in (see "Sharing and hand-off"):

```
Implement the R and Python cleaners from the checked-in contract. Add tests
for missing values, unknown flags, malformed dates, and cross-language
agreement. Use `just qa` (defined below) and stop if the contract is ambiguous.
```

The model drafts the code. The fixture, expected output, container build, and comparison decide whether it is correct. Pi also writes some tests, so review them: a test written to match the model's own output proves nothing.

### 11.3 Build a Digest-Pinned Rocker Image

Rocker's versioned tags are not immutable: the images are rebuilt periodically, so `4.6.1` today is not byte-identical to `4.6.1` last month (Rocker's reproducibility guide says to use a digest if you need the same image). Pin the multi-architecture image index digest, so the pin works on amd64 and arm64. The value below was current on 24 September 2026 and will be stale after the next rebuild; look up the current one first. `podman image inspect` reports a single-platform manifest digest, which is the wrong source.

```
skopeo inspect --format '{{.Digest}}' docker://ghcr.io/rocker-org/tidyverse:4.6.1
```

```
ARG ROCKER_IMAGE=ghcr.io/rocker-org/tidyverse:4.6.1
ARG ROCKER_DIGEST=sha256:94a4cdb7be4d1214f9039bca4937e8919253d2ad90d8bed084e9a8af685a50c7
FROM ${ROCKER_IMAGE}@${ROCKER_DIGEST}

USER root
# python3 is already in the image; the venv module is not.
RUN apt-get update \
 && apt-get install -y --no-install-recommends python3-venv \
 && rm -rf /var/lib/apt/lists/* \
 && python3 -m venv /opt/venv

ENV PATH="/opt/venv/bin:${PATH}"
WORKDIR /work

COPY requirements.txt .
# --require-hashes means every requirement must be pinned with a hash.
RUN pip install --no-cache-dir --require-hashes -r requirements.txt

COPY renv.lock .
RUN R -q -e 'install.packages("renv")' \
 && R -q -e 'renv::restore(prompt = FALSE)' \
 && rm -rf /tmp/downloaded_packages

USER rstudio
```

The virtual environment isolates the pinned Python packages from the Ubuntu system Python. Tag pinning alone does not make the R packages reproducible, because Rocker installs the selected R version from a moving CRAN snapshot; that is why the image uses `renv`. Four details matter:

- `renv::restore()` in a project that is not activated installs into the image's system library and can overwrite Rocker's own packages. An incomplete lockfile, or one taken from another machine, can downgrade packages that the tidyverse relies on, and the image then fails at load time. Generate `renv.lock` from a container running this same image, so it records Rocker's package set as well as yours:

```
podman run --rm -v "$PWD:/w:Z" -w /w ghcr.io/rocker-org/tidyverse:4.6.1 \
  R -q -e 'install.packages("renv"); renv::snapshot(type = "all", lockfile = "renv.lock",
  prompt = FALSE)'
```

- The bootstrap installs whatever `renv` version the mirror serves that day; the lockfile records `renv`'s version but this step does not honour it.
- `requirements.txt` must carry hashes. Write direct dependencies (here `pytest`) in `requirements.in` and run `pip-compile --generate-hashes requirements.in`, or `uv pip compile --generate-hashes requirements.in -o requirements.txt`.
- Under `--userns=keep-id` (below) the process runs as your host user whatever `USER` says; `USER rstudio` (UID 1000 in this image) serves runtimes without keep-id. Use `.containerignore` as an allow-list (`*`, then `!requirements.txt` and `!renv.lock`), because the build context includes everything not ignored.

### 11.4 Run It With Rootless Podman

A small `Justfile` exposes the workflow without hiding the container boundary. These recipes were built and run against a toy implementation with Podman 5.7.0 on Ubuntu, and the toy project passed `just qa`:

```
set shell := ["bash", "-ceu"]

IMAGE   := "localhost/pi-temperature:dev"
ROOT    := justfile_directory()
# Podman already mounts a tmpfs on /tmp for read-only containers; --tmpfs adds a size cap.
SECURE  := "--rm --userns=keep-id --cap-drop=all --security-opt=no-new-privileges " + \
           "--read-only --network=none --tmpfs /tmp:rw,size=256m"
ENV     := "-e PYTHONDONTWRITEBYTECODE=1 -w /work"
# Read-only inputs: only the directories the container needs, not the repository root.
INPUTS  := "-v " + ROOT + "/r:/work/r:ro,Z -v " + ROOT + "/python:/work/python:ro,Z " + \
           "-v " + ROOT + "/scripts:/work/scripts:ro,Z -v " + ROOT + "/data:/work/data:ro,Z"
OUTPUTS := "-v " + ROOT + "/outputs:/work/outputs:Z"
RUN     := "podman run " + SECURE + " " + ENV + " " + INPUTS

build:
  podman build -t {{IMAGE}} -f Containerfile .

test-r:
```

```
  {{RUN}} {{IMAGE}} Rscript -e 'testthat::test_dir("r/testthat")'

test-python:
  {{RUN}} {{IMAGE}} pytest -q -p no:cacheprovider python/tests

pipeline:
  mkdir -p outputs
  {{RUN}} {{OUTPUTS}} {{IMAGE}} Rscript r/clean.R
  {{RUN}} {{OUTPUTS}} {{IMAGE}} python python/clean.py
  {{RUN}} {{OUTPUTS}} {{IMAGE}} python scripts/compare_outputs.py

qa: test-r test-python pipeline
```

Only `outputs/` is writable, so a buggy or malicious script cannot alter the fixture or expected output. Mounting only the needed sub-directories keeps `.git/config` (which may hold remote URLs or tokens) and `.git/hooks` out of the container, and matters because `:Z` relabels the mounted tree for SELinux hosts (and is ignored elsewhere). `outputs/` must exist before Podman starts, since Podman refuses a bind mount with a missing source, so `pipeline` creates it. `-p no:cacheprovider` and `PYTHONDONTWRITEBYTECODE` stop pytest and Python writing into read-only mounts, and `--network=none` means the runs cannot fetch anything.

The recipes call `podman` directly because `--userns=keep-id` is Podman-only. On Docker, use `--user "$(id -u):$(id -g)"` instead and drop `:Z` unless the host uses SELinux. Never mount the repository parent, home directory, SSH agent, cloud credentials, or a container socket.

### 11.5 Verify the Result Independently

Pi should run `just qa` and report its output, but its report of its own run is not evidence. Run `just qa` yourself and check:

1. Both test suites pass, and unknown flags and malformed dates fail as expected.
2. The `missing` rows do not contribute to `n_valid` or the statistics, and the `2026-01-02` station-day produces no row.
3. The R and Python outputs match the expected file within the declared tolerance (`compare_outputs.py` checks this).
4. `git diff --stat` shows only the intended source, lock, and test changes: the `Justfile`, `Containerfile`, `.containerignore`, `AGENTS.md`, and the fixture and expected files are unchanged, and nothing was written into the source tree.

Item 4 matters most. Pi runs on the host with no approval prompts, so it can edit the `Justfile` or `Containerfile` to loosen `--network=none`, `--read-only`, or the mounts. The container flags are a convention Pi follows, not a boundary it is subject to. Commit those files first and review any change to them as a separate diff.

If the project later uses real data, keep it outside Git and mount it read-only. Do not give Pi access to sensitive datasets or credentials merely because the analysis container can read them. This example teaches reproducibility and review; it does not make Pi a trusted data-processing authority.

# 12 Reference

Look-up material for when something breaks, when you need to find a file, and when you want the next document to read.

## 12.1 Troubleshooting Order

Work from the process outwards: each step only makes sense if the previous one is green.

1. Executable: `command -v pi`, `pi --version`, `pi --help`.
2. Working directory and discovered context files: the startup header, `--verbose`, and `-nc` (`--no-context-files`) to exclude `AGENTS.md` and `CLAUDE.md`. If a project skill or extension is missing, check project trust (`-a` or `-na` forces a decision for one run).
3. Provider authentication, without printing secrets: `pi auth check --provider <name>`. Avoid `pi auth print-api-key` and `print-bearer-token`, which write secrets to stdout.
4. Model response and tool-call support.
5. Provider or search network path. `--offline` excludes catalogue refresh.
6. Underlying `curl`, `gh`, browser helper, or local server.
7. Extension or skill disabled and retested: `--no-extensions` (`-ne`), `--no-skills` (`-ns`), `--no-prompt-templates` (`-np`), or `-nt` for no tools; `pi config` toggles package resources.
8. Fresh or `--no-session` run to exclude stale context.
9. Independent reproduction of the repository failure.

Do not install another extension until you know which layer failed.

## 12.2 Configuration Map

```
~/.pi/agent/                    agent directory (override: PI_CODING_AGENT_DIR)
  settings.json                 user settings, defaults, package declarations
  models.json                   custom providers and models
  auth.json                     saved API keys and OAuth credentials (protect it)
  trust.json                    saved project-trust decisions
  sessions/                     session files, grouped by working directory
  extensions/  skills/  prompts/  themes/
  AGENTS.md | CLAUDE.md         instructions applied in every directory
  SYSTEM.md, APPEND_SYSTEM.md   replace or extend the system prompt
~/.agents/skills/               portable Agent Skills location (also read by Pi)
```

```
.pi/                            project settings and resources (load after project trust)
  settings.json  extensions/  skills/  prompts/  themes/  SYSTEM.md  APPEND_SYSTEM.md
.agents/skills/                 portable project skills (also trust-gated)
AGENTS.md | CLAUDE.md           project instructions (loaded regardless of trust)
```

## 12.3 Recommended Starting Stack

This is an order of adoption, not a day-one install. In the first week, take items 1, 2, and 4; add the rest when a concrete need appears.

1. The current Pi installer or npm package.
2. One hosted tool-capable provider.
3. One local OpenAI-compatible server, tested separately.
4. One small project skill for repository procedures.
5. A maintained search skill with a dedicated API key.
6. Browser tooling, only when static search is insufficient.
7. A container or VM for untrusted or unattended work. A disposable Git worktree limits accidental edits, not what Pi can access.

Add one capability at a time, test it, record its permissions, and keep it only if it solves a recurring problem.

## 12.4 Further Reading

### Pi documentation

- Pi repository, documentation index, quickstart, and how Pi works: start here.
- Pi changelog and 1.0.4 release: the October 2026 baseline.
- Built-in MCP and codemode: server configuration, exposure, and tool-call composition.
- Security and isolated environments: project trust, and Docker, Docker Sandboxes, OpenShell, and Gondolin.
- Command line, configuration, sessions, and slash commands: flags, the agent directory, branching, and compaction.
- Models, providers, and llama.cpp: authentication, `models.json`, and the router integration.
- Skills, prompt templates, extensions, packages, and themes.
- CLI integration, RPC, and the SDK: print, JSON, RPC, and embedding.
- Package catalogue and model catalogue; extension examples; releases.

### Design and ecosystem

- Mario Zechner, What I learned building an opinionated and minimal coding agent: the design rationale behind everything in "Why Pi is built this way".
- Mario Zechner, I've sold out: the move to Earendil and the licensing plan.
- `pi-skills`, `pi-web-access`, `pi-mcp-adapter`, and the `oh-my-pi` fork.
- Agent Skills specification, the format shared with Claude Code.
- Radius, Earendil's hosted gateway for Pi.

**Containers and reproducibility**

- Podman run and Podman build references.
- Rocker: versioned images, reproducibility, rootless Podman, and extending images.
- `renv` and Docker, pip hash-checking, and the `just` manual.

**Other agents and related primers**

- Aider, OpenCode, Claude Code, and Codex CLI.
- Claude Code alternatives primer, Codex for Claude Code users primer, and OpenClaw primer.

# 13 Closing Perspective

Pi's distinctive feature is its programmable loop and selective tool exposure. Built-in MCP and optional codemode now supply tool integration and composition; you still decide which browser, search service, approval policy, planning workflow, and subagent strategy belong around it.

Pi suits you if you want a loop you can program: a provider-neutral harness, an embeddable runtime, a research workflow you assemble yourself. It suits you less if you want bundled per-tool approvals, subagents, and web search on day one. MCP is now supplied, but its servers still need configuration and review. Pi runs with your permissions, so put a container or VM around anything untrusted or unattended, and review extensions, packages, and MCP servers as code you are about to execute.

## 13.1 Where to Go Next

A sensible first week: install Pi and check `pi --version`; run the read-only orientation from "A controlled first session" on a repository you know; add one skill; then repeat the Rocker example with your own fixture. Read the author's design essay before deciding what else to add.